

Module 08: Trivial File Transfer Protocol (TFTP) vs FTP and HTTP

Gateway Classes Lecture Series | Dr. Nidhi Parashar Ma'am | Author: Shivam Keshari

TRIVIAL FTP (TFTP) vs FTP vs HTTP ARCHITECTURAL MATRIX

UDP Bootstrapping Mechanics, 512-Byte Blocks, Stop-and-Wait ACK & Protocol Comparison

TFTP Architecture (RFC 1350 — UDP Port 69)

Transport Protocol: UDP Port 69 (Minimalist, connectionless). • **No Authentication:** Zero username / password security check. • **Fixed 512-Byte Data Blocks:** Data split into fixed 512-byte payload. • **Stop-and-Wait Error Control:** Sender sends Block N, waits for ACK N. • **Termination Signal:** Block size < 512 bytes signals EOF (End of File). • **Primary Use-Case:** PXE Network Booting, loading Cisco IOS firmware. 5 Packet Types: RRQ (1), WRQ (2), DATA (3), ACK (4), ERROR (5) TFTP STOP-AND-WAIT INTERACTION Client (TID) Server (69) RRQ (Read Request "firmware.bin") DATA Block 1 (512 Bytes) ACK Block 1 DATA Block 2 (214 Bytes < 512 = EOF) 3-WAY MASTER ARCHITECTURAL COMPARISON: FTP vs TFTP vs HTTP FTP (RFC 959) • **Transport:** TCP Ports 20 & 21 • **Connections:** Dual (Control & Data) • **Security:** User/Pass auth support • **Directories:** Full browsing, rename, delete • **Memory Footprint:** Heavy implementation TFTP (RFC 1350) • **Transport:** UDP Port 69 • **Connections:** Connectionless datagrams • **Security:** None (No auth / cleartext) • **Directories:** Cannot list or change dirs • **Footprint:** Ultra-light (Fits in ROM/BIOS) HTTP (RFC 2616) • **Transport:** TCP Port 80 / 443 • **Connections:** Single TCP stream per session • **Security:** HTTPS SSL/TLS encryption • **Primary Use:** Hypertext Web & REST APIs • **Footprint:** Rich feature stack & caching

1. Trivial FTP Architecture & Minimalist Design

TFTP is an ultra-lightweight, UDP-based file transfer protocol engineered to fit into low-memory bootstrap ROMs and BIOS chips.

The Stop-and-Wait Transfer Engine

Operating over **UDP Port 69**, TFTP divides data into fixed 512-byte blocks. Each transmitted block demands explicit acknowledgment before the succeeding block can be released. A block < 512 bytes signals End of File (EOF).

2. Master 3-Way Comparative Matrix

Property	FTP (RFC 959)	TFTP (RFC 1350)	HTTP (RFC 2616)
Transport Protocol	TCP (Port 20 Data, Port 21 Control)	UDP (Port 69)	TCP (Port 80 / 443)
Authentication	Supported (USER / PASS)	None (Zero credentials)	Headers, Basic, Bearer, OAuth
Directory Navigation	Supported (LIST, PWD, CWD)	Not Supported	URL path routing
Application Niche	Enterprise file management	Router firmware loading, PXE boot	Web content delivery & REST APIs